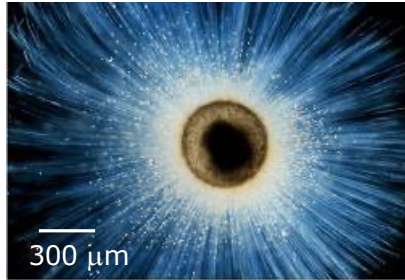


Living

*Orbulina
universa*



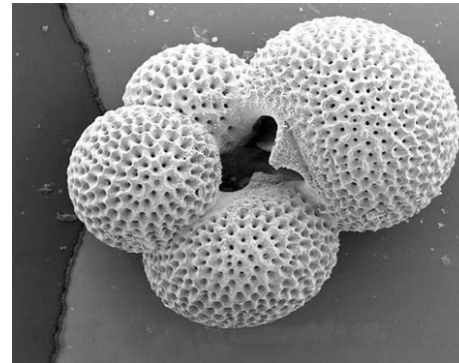
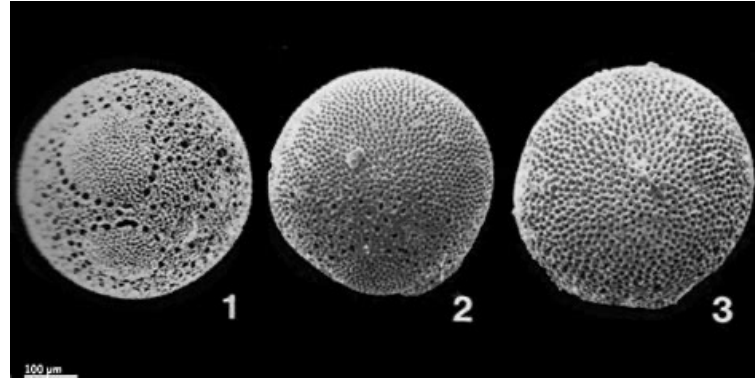
*Globigerinoides
ruber*

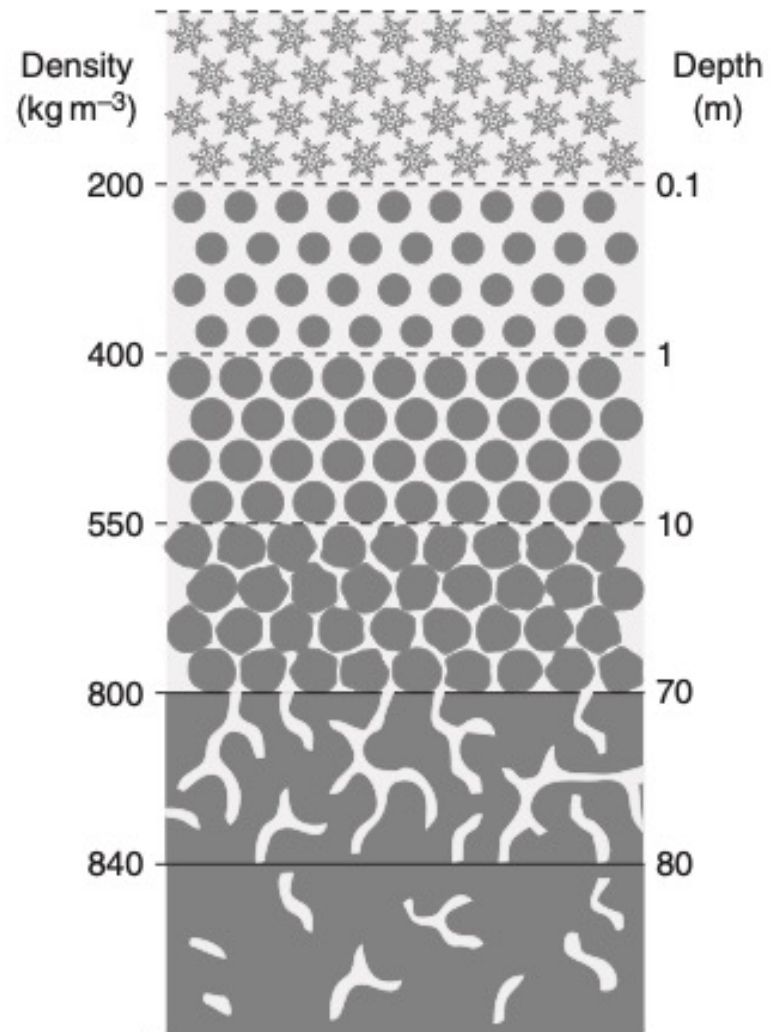


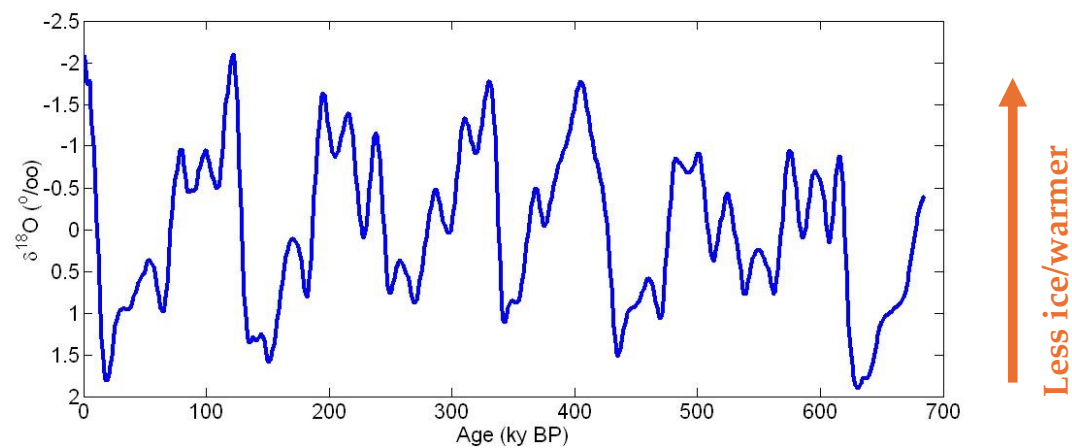
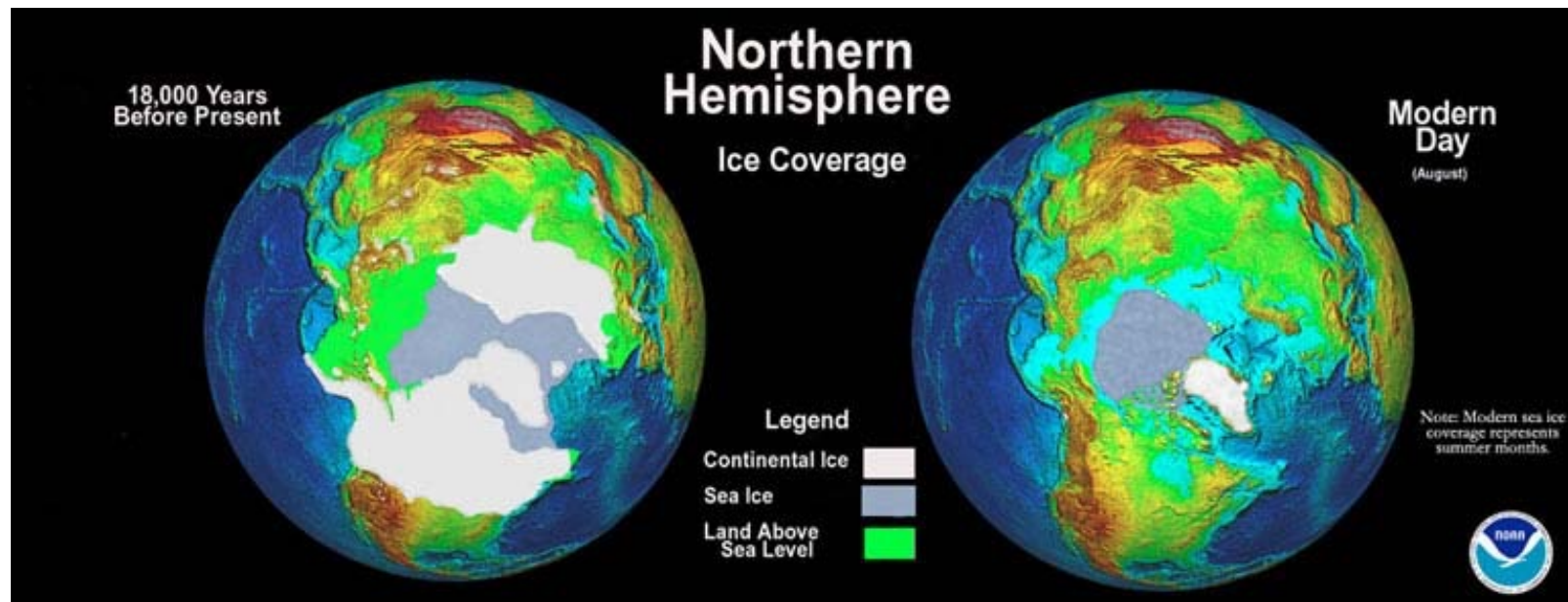
*Globigerina
bulloides*

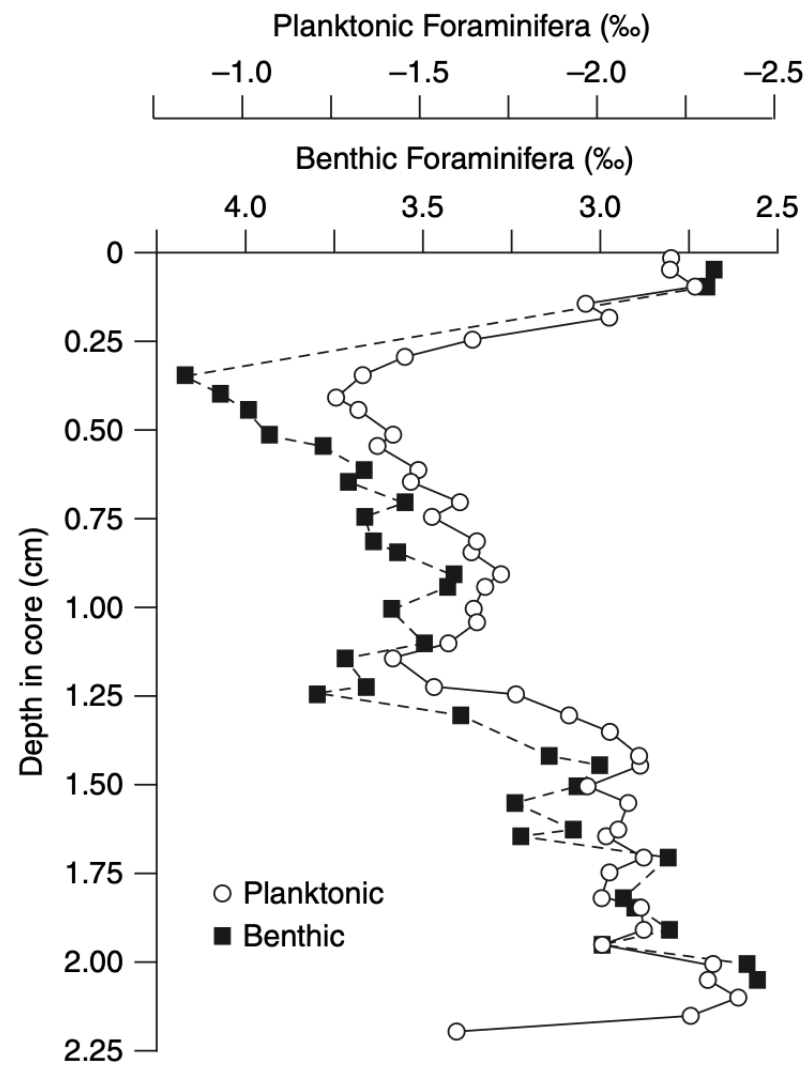


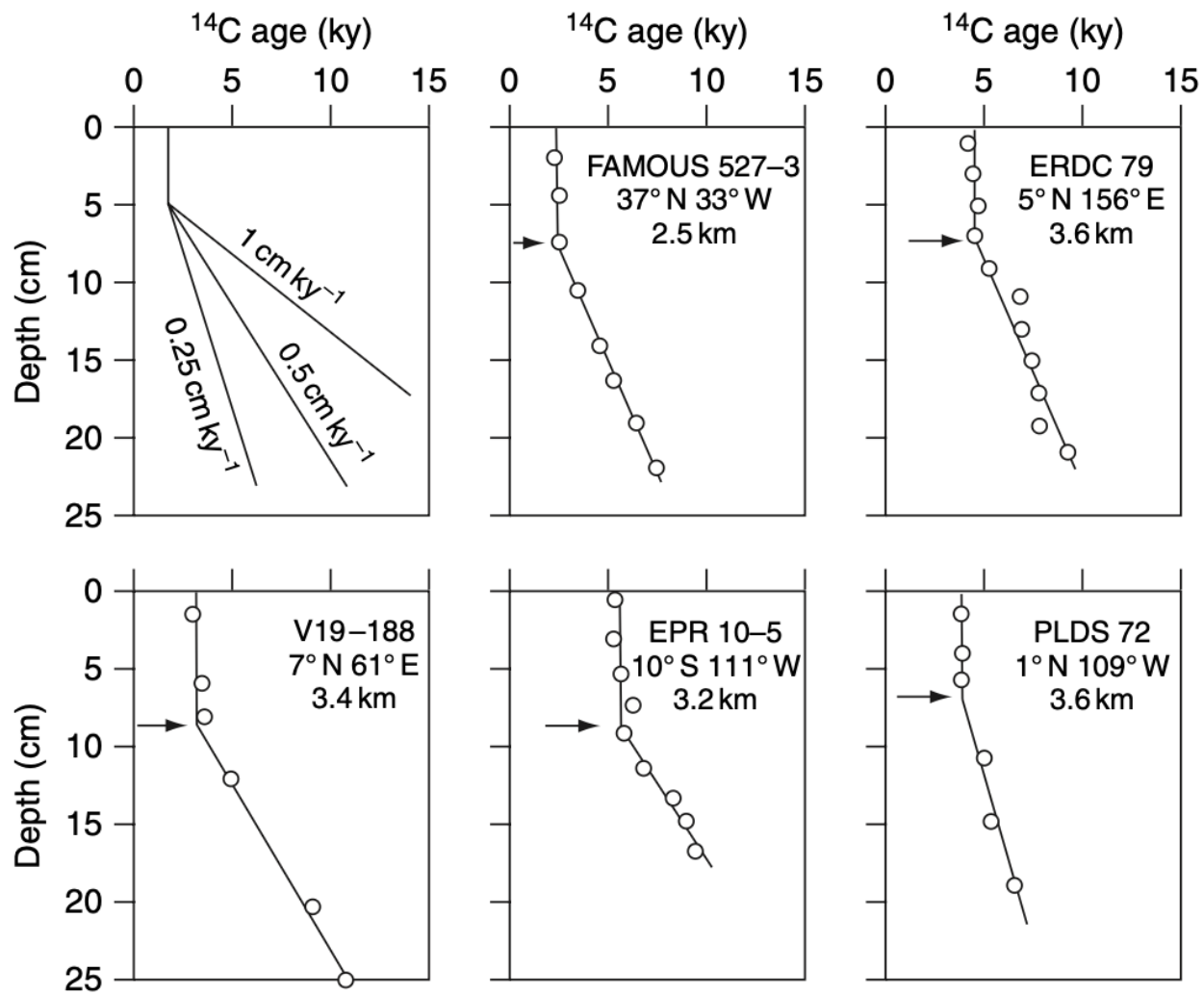
Spero (1998)

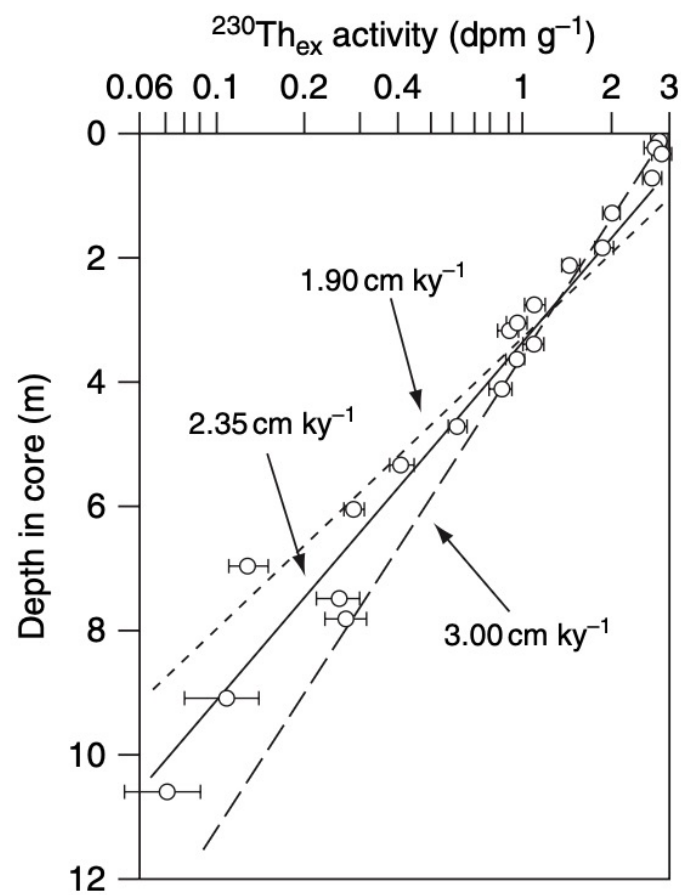
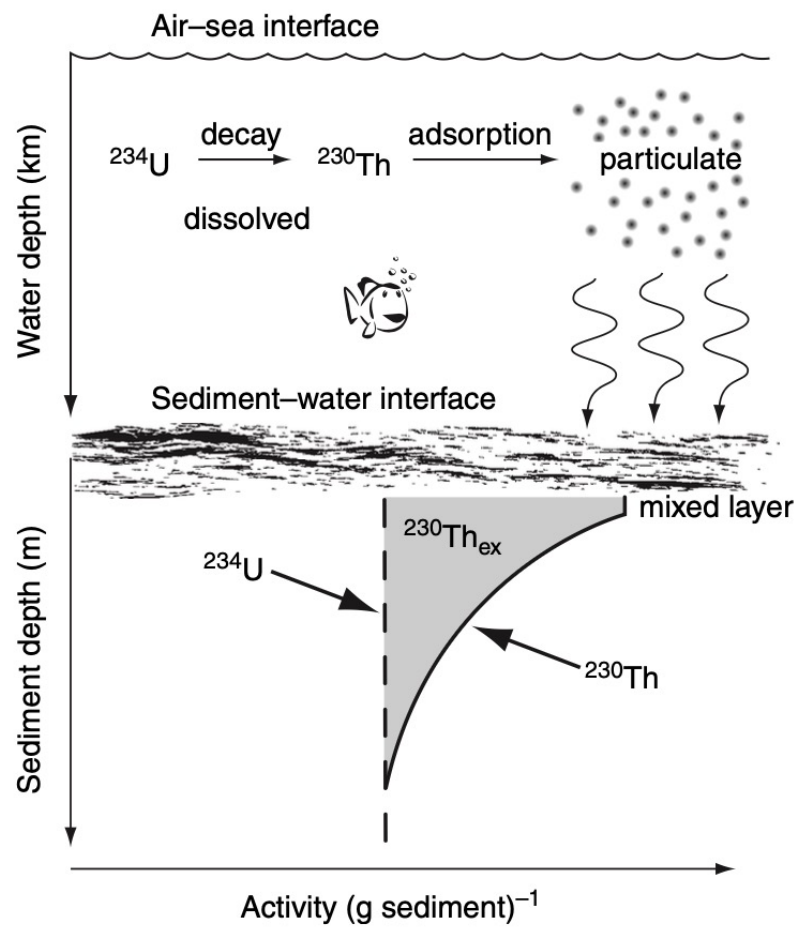






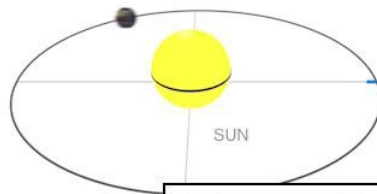






Orbital Cycles

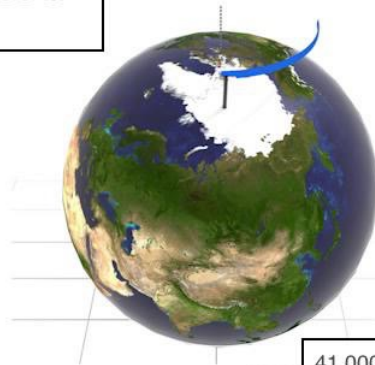
Three periodic motions in Earth's orbit, known as Milankovitch cycles, contribute a predictable amount of variation to Earth's climate over time frames of tens of thousands to hundreds of thousands of years.



100,000-year cycles
Changes in Eccentricity
(Orbit Shape)

*Changes in eccentricity exaggerated so the effect can be seen. Earth's orbit shape varies between 0.0034 (almost a perfect circle) to 0.058 (slightly elliptical).

26,000-year cycles
Axial Precession
(Wobble)



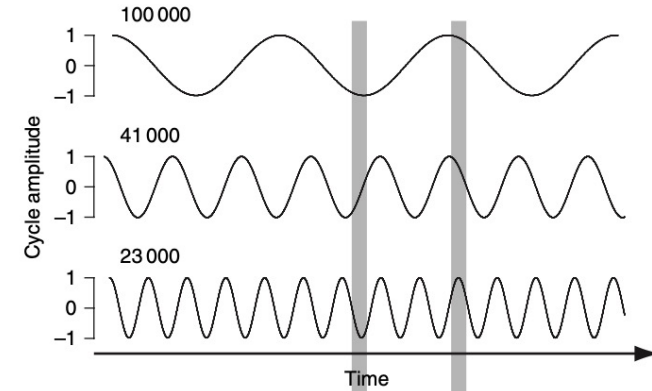
41,000-year cycles
Changes in Obliquity
(Tilt)

Relative to orbital plane
22.1-24.5°



climate.nasa.gov

(A) Individual sine-wave cycles



(B) Combinations of cycles

